

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-305 — Network Configuration and Management

Time: Three hours

Maximum Marks: 70

Instructions: Answer all sections. Candidates are required to write their answers in their own words as far as practicable. Symbols have their usual meanings.

SECTION — A

Long Answer questions. Answer two questions, selecting one from each pair of OR questions:
2 × 15 = 30 Marks

Question 1. Describe the OSI model with a suitable diagram and compare the OSI model with the TCP/IP model. **OR**

Explain the process of inter-VLAN network communication. What do you understand by a Layer 3 switch?

Question 2. What is the IEEE 802.11 protocol? Explain the 802.11 infrastructure network architecture and its components. **OR**

Discuss any three Internet applications with illustrative diagrams.

SECTION — B

Question 3. Attempt any six parts (Answer in about 200 words each): **6 × 5 = 30 Marks**

- (a) What are the various network topologies? Discuss briefly.
- (b) Describe Repeater, Hub, and Switch. Compare a Switch with a Bridge.
- (c) Discuss the IEEE 802.3 Ethernet protocol and its different categories.
- (d) What do you understand by a Distribution System (DS) in wireless networking?
- (e) What is an Access Point (AP)? What are its various deployment methods?
- (f) Explain the BGP routing protocol and discuss its two types.
- (g) What is a DNS server? Explain its working with a suitable diagram.
- (h) How is Wireshark helpful in network troubleshooting and packets analysis?

SECTION — C

Question 4. Attempt all parts:

10 × 1 = 10 Marks

- (a) UTP stands for _____.
- (b) In network models, cables are considered to function at Layer _____.

- (c) In network models, routers are considered to operate at Layer _____.
- (d) The WLC and APs use the _____ protocol to communicate.
- (e) WPA stands for _____.
- (f) The broadcasting address of a network like 192.168.1.0/24 is 192.168.1._____.
- (g) Subnetting is the short form of _____.
- (h) Is UTM a firewall or not?
- (i) Which protocol is more reliable for data transmission — TCP or UDP?
- (j) Which command/utility is used to trace the routing path of a packet?